
BRYCE REID

ELECTRICAL ENGINEER

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SUMMARY

- Recent graduate of the Electrical Engineering bachelor's program at British Columbia Institute of Technology (BCIT)
- Strong foundation in core engineering disciplines including circuit design, power systems, and process control
- Primary interests include embedded systems, RTL design, and DSP
- I enjoy working at the intersection of hardware and software to translate high-level concepts into practical, low-level implementations
- Seeking opportunities in embedded systems, digital / RTL / FPGA / ASIC design, or design verification
- Open to roles across the broader electrical engineering domain

SKILLS

PROGRAMMING | C/C++ | Python | SystemVerilog | HTML | JavaScript | TCL |

EMBEDDED SYSTEMS | ARM Microcontrollers | FPGA | SPI | I2C | UART | I2S | FreeRTOS | Zephyr RTOS | Bluetooth LE | OpenCV |

DSP | Fourier Analysis | Z-Transforms | IIR/FIR Filters | Spatial Filtering | DCT | Wavelet Transformations |

SOFTWARE SYSTEMS | Agile/Scrum | CI/CD | Docker | REST API | Firestore | Firebase Auth | Pytest | Cloud Deployment |

SOFTWARE TOOLS | VS Code | MATLAB | Simulink | LTSpice | ModelSim | Git | AutoCAD | Solidworks | KiCAD | Microsoft Office |

PROJECTS

WIRELESS WEARABLE GROUP INTERCOM

- Engineered a custom Bluetooth LE Audio star topology that achieved reliable wireless audio streaming across 4 devices at up to 18 m, replacing an unsuitable Bluetooth Mesh proposal by designing unicast audio routing and active speaker selection modules in C using the nRF Connect SDK and Zephyr RTOS
- Led a 4-person firmware team as Team Lead, coordinating workplans tailored to individual strengths, delivering bi-weekly stakeholder reports, and running training sessions to onboard team members on an unfamiliar toolchain
- Eliminated cellphone dependency for group cyclist communication by selecting Bluetooth LE Audio over cellular and Wi-Fi alternatives, meeting the industry sponsor's core standalone-operation requirement
- Solved a many-to-many audio connectivity gap with no existing reference implementations by architecting a push-to-talk protocol that minimizes bandwidth and prevents crosstalk, ensuring only one audio stream transmits at a time across the intercom network

LFO GENERATOR

- Delivered a fully functional variable-waveshape LFO signal generator on an FPGA by implementing a modular SystemVerilog design with components for SPI-driven DAC output, lookup-table waveform synthesis, clock generation, and rotary encoder input handling
- Accelerated simulation workflows by authoring TCL scripts in ModelSim to automate compilation, simulation execution, and waveform setup, reducing manual effort across iterative debug cycles
- Validated design correctness at the module level by writing dedicated SystemVerilog testbenches and using simulation tools to surface and resolve functional issues before hardware integration

MIDI DRUM SEQUENCER

- Built a functional 4-track, 16-step MIDI drum sequencer on a Raspberry Pi Pico 2 W by implementing bidirectional USB MIDI via TinyUSB for DAW-synchronized playback and sound triggering in C
- Met real-time sequencer timing constraints on a bare-metal microcontroller by architecting the firmware around FreeRTOS for deterministic task scheduling

BREATHING CUBE WEB APP

- Delivered a production-ready distributed backend for an IoT meditation habit tracker in an Agile Scrum team by authoring REST API and web routes in Python, with Firestore database architecture and Firebase JWT and API key authentication
- Accelerated hardware-independent development by building a GUI-based ESP-32 simulator in Python that replicated cube start/stop/reset commands and displayed live JSON server responses for API testing
- Enforced code quality across the team by implementing a Pytest suite with mocked client, Firebase Auth, and Firestore dependencies, integrated into a CI pipeline to gate all pull requests on passing tests

AUTOMATED ROBOT CAR

- Delivered a fully autonomous robot car that completed a multi-target navigation course in under 40 seconds, earning full marks, by sole-authoring all software in C++ on a Raspberry Pi 4 running Linux
- Implemented real-time computer vision localization using OpenCV to detect a QR code mounted on the car from an overhead video feed, enabling precise position tracking and autonomous course navigation
- Coordinated targeting, projectile firing, and autonomous parking through a single C++ software stack integrating serial communication, GPIO hardware control, and socket-based networking

COLOR-BASED SORTING MACHINE

- Achieved 4-bin color-based sorting of pink, green, blue, and yellow balls by implementing HSV color segmentation and contour detection in C++ with OpenCV on a Raspberry Pi, exceeding the 2-bin baseline requirement
- Designed and built a custom mechanical sorting apparatus featuring a center-loaded vertical ball chute, dual servo-controlled ramps, and four corner bins contained in a single box, enabling fully automated physical sorting driven by computer vision output
- Delivered a complete hardware-software integrated system with GPIO-controlled servos, physical pushbuttons for manual override and start/stop control, and LED bin indicators, implemented using object-oriented design principles in C++

SMALL PERSONAL PROJECTS

- Built multiple DIY guitar/bass effects pedals - this involves reading schematics, soldering PCBs, signal testing, and electronic assembly
- Built an analog kick drum synthesizer circuit on a breadboard

EDUCATION

BACHELOR OF ENGINEERING IN ELECTRICAL ENGINEERING

British Columbia Institute of Technology

2021 – 2026

Burnaby, BC

EMPLOYMENT

PROJECT ASSISTANT, CO-OP

Houle Electric

May 2024 – Sep 2024

North Vancouver, BC

- Improved project visibility for the management team by conducting independent on-site field reviews of electrical installations and maintaining up-to-date drawing markups for progress tracking
- Reduced trade conflicts on the jobsite by logging constraints blocking Houle field workers and authoring a weekly constraints report distributed to the general contractor
- Owned constraint resolution end-to-end by coordinating correction plans directly with general contractor representatives and following up on outstanding deadlines on behalf of Houle Electric